

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Elk Grove Village, Illinois -- station KORD, America/Chicago
Committed pair OSM 863162820 -> 377032061
Stream Chicago II / Equinix Chicago CH3
Facade gap 118.4 m between the two halls
Weather record 43,775 real hourly records from KORD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4108 C at 240 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-08-30 (recirc_edge)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 2 mode changes. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.215	21.0	17.780	1.071
01	FREE	yes	-	18.197	21.0	17.780	1.005
02	FREE	yes	-	18.199	21.0	17.780	0.970
03	FREE	yes	-	18.228	21.0	17.780	0.935
04	FREE	yes	-	18.245	21.0	17.780	0.895
05	FREE	yes	-	18.265	21.0	17.780	0.820
06	FREE	yes	-	18.725	21.0	18.330	0.770

07	FREE	yes	-	20.019	21.0	18.330	1.468
08	mechanical	no	dry-bulb	21.176	21.0	18.890	1.731
09	mechanical	no	dry-bulb	22.352	21.0	19.440	1.842
10	mechanical	no	dry-bulb	22.965	21.0	20.560	1.779
11	mechanical	no	dry-bulb	22.874	21.0	21.110	1.325
12	mechanical	no	dry-bulb	23.158	21.0	21.670	1.366
13	mechanical	no	dry-bulb	24.010	21.0	22.613	1.336
14	mechanical	no	dry-bulb	23.192	21.0	22.220	0.982
15	mechanical	no	dry-bulb	22.967	21.0	21.670	1.108
16	mechanical	no	dry-bulb	22.639	21.0	21.110	1.152
17	mechanical	no	dry-bulb	21.791	21.0	20.560	0.952
18	mechanical	yes	switch budget	20.952	21.0	20.000	0.982
19	mechanical	yes	switch budget	20.098	21.0	18.890	1.173
20	mechanical	yes	switch budget	19.634	21.0	18.330	1.285
21	mechanical	yes	switch budget	18.968	21.0	17.780	1.042
22	mechanical	yes	switch budget	18.416	21.0	17.780	0.882
23	mechanical	yes	switch budget	18.178	21.0	17.220	1.081

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.215 C, which is 2.785 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.071 C, and the margin is measured, not chosen: 0.912 C of group-conditional forecast error for this hour of day, plus 0.1599 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.197 C, which is 2.803 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.005 C, and the margin is measured, not chosen: 0.863 C of group-conditional forecast error for this hour of day, plus 0.1418 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.199 C, which is 2.801 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.970 C, and the margin is measured, not chosen: 0.825 C of group-conditional forecast error for this hour of day, plus 0.1442 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.228 C, which is 2.772 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.935 C, and the margin is measured, not chosen: 0.762 C of group-conditional forecast error for this hour of day, plus 0.1729 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.245 C, which is 2.755 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.895 C, and the margin is measured, not chosen: 0.710 C of group-conditional forecast error for this hour of day, plus 0.1853 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.265 C, which is 2.735 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.820 C, and the margin is measured, not chosen: 0.615 C of group-conditional forecast error for

this hour of day, plus 0.2052 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.725 C, which is 2.275 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.770 C, and the margin is measured, not chosen: 0.661 C of group-conditional forecast error for this hour of day, plus 0.1097 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.019 C, which is 0.981 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.467 C, and the margin is measured, not chosen: 1.225 C of group-conditional forecast error for this hour of day, plus 0.2424 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.176 C against a 21.0 C limit, so it fails by 0.176 C. It would take a limit 0.176 C higher, or a bound 0.176 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.352 C against a 21.0 C limit, so it fails by 1.352 C. It would take a limit 1.352 C higher, or a bound 1.352 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.965 C against a 21.0 C limit, so it fails by 1.965 C. It would take a limit 1.965 C higher, or a bound 1.965 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.874 C against a 21.0 C limit, so it fails by 1.874 C. It would take a limit 1.874 C higher, or a bound 1.874 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.158 C against a 21.0 C limit, so it fails by 2.158 C. It would take a limit 2.158 C higher, or a bound 2.158 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.010 C against a 21.0 C limit, so it fails by 3.010 C. It would take a limit 3.010 C higher, or a bound 3.010 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.192 C against a 21.0 C limit, so it fails by 2.192 C. It would take a limit 2.192 C higher, or a bound 2.192 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.967 C against a 21.0 C limit, so it fails by 1.967 C. It would take a limit 1.967 C higher, or a bound 1.967 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.639 C against a 21.0 C limit, so it fails by 1.639 C. It would take a limit 1.639 C higher, or a bound 1.639 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.791 C against a 21.0 C limit, so it fails by 0.791 C. It would take a limit 0.791 C higher, or a bound 0.791 C tighter, to change this hour.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.952 C

against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.098 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.634 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.968 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.417 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.178 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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